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Low Data Rate Receiver with Inbuilt Modulator

Low data Rate Receiver modem is a high performing system for receiving & processing satellite or aircraft data. It is deployed most commonly within a satellite ground station with data reception and processing at high-rate signals from LEO or GEO Satellites. It is configurable to accommodate multiple demodulation schemes & decoding algorithms. It is suitable for low data rate scientific, remote sensing and telecommunications applications.

Introduction

The Low Data Rate Receiver with Inbuilt Modulator (LDR) is a high-performance communication system designed for reliable reception and processing of satellite and aircraft data signals. It is primarily deployed in satellite ground stations to receive and process high-rate signals from Low Earth Orbit (LEO) and Geostationary Earth Orbit (GEO) satellites. The system is highly configurable, supporting multiple demodulation schemes, decoding algorithms, and data formats, making it suitable for a wide range of applications including scientific missions, remote sensing, and telecommunications.

Built on an advanced FPGA-based architecture, the LDR integrates complete baseband signal processing along with an inbuilt modulator and channel emulation capabilities. This enables efficient handling of low data rate signals, flexible system configuration, and reliable performance under varying operational conditions. With its fully digital, tunable receiver and transmitter design, along with support for modern monitoring, control, and interface options, the LDR provides a comprehensive and scalable solution for satellite communication and data reception systems.

Architecture

The architecture of the Low Data Rate Receiver with Inbuilt Modulator (LDR) is based on a fully digital, FPGA-centric design that integrates all baseband signal processing functions into a single platform. The system consists of an IF receiver, demodulator, decoding unit, and an inbuilt modulator with channel emulation capability, enabling both signal reception and transmission within the same unit. It supports one or two IF input channels at 70 MHz and processes signals through configurable demodulation schemes such as BPSK and QPSK, followed by advanced decoding techniques including Viterbi and differential decoding. The architecture also includes flexible input/output interfaces such as LVDS, ECL, and Ethernet, along with real-time monitoring, control, and configuration through local or remote interfaces. This integrated and modular design ensures high performance, adaptability, and efficient handling of low data rate satellite communication signals.

Why Choose Us?

🚀 Rapid integration, faster time-to-market

📘 Clean, verified, documented RTL

🤝 Direct access to experts

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Features

- Fully digital, FPGA-based architecture with complete baseband processing integrated into a single platform.
- Supports reception and processing of satellite and aircraft signals from LEO and GEO systems.
- Configurable demodulation schemes including BPSK and QPSK.
- Supports advanced decoding techniques such as Viterbi decoding and differential decoding.
- Operates at 70 MHz IF with symbol rates from 20 KSps to 20 MSps.
- Handles wide input signal power range from -5 dBm to -50 dBm.
- Provides 1 or 2 IF channels with flexible configuration options.
- Includes an inbuilt modulator with channel emulation capabilities such as noise, Doppler, and path loss simulation.
- Supports multiple data formats and output interfaces including LVDS, ECL, and Ethernet.
- Offers real-time monitoring of parameters like RSSI, SNR, BER, and constellation display.
- Enables local and remote configuration through front panel GUI and TCP/IP-based control.
- Compliant with CCSDS recommendations for satellite communication systems.

IF Receiver and Demodulator Specifications

IF INPUT	
Parameter	**Specification**
No of IF Inputs	• 1 or 2 (Optional)
Input Frequency	• 70MHz
Max Symbol Rate @ 70MHz	• 20MSps
IF Tunability	• ±10 KHz step size
Power	• -5dBm to -50 dBm
VSWR	• <= 1.5
IF Connector Type	• SMA Female
Acquisition Range	• Programmable Up-to ±300 KHz/Auto
Doppler Rate	• 2KHz/sec
Acquisition Time	• <500ms (higher for lower data-rate)
IF Connector Type	• SMA Female (50 ohm)

DEMODULATION & BIT-SYNC

Parameter	Specification
Supported schemes	• BPSK, QPSK (Selectable)
Spectral Inversion support	• I&Q Swap, $\bar{I}$&$\bar{Q}$, $\bar{I}$&Q, I&$\bar{Q}$
Symbol Rate	• 20KSps - 20 MSPS
Max Bit rate	• BPSK (1*20) Mbps
	• QPSK (2*20) Mbps
BER Degradation	• BPSK/QPSK: 1dB @10^{-6} BER
Phase Ambiguity Resolution	• Automatic or Manual

DECODING

Parameter	Specification
Viterbi Decoding	• Selectable Rate 1/2 Punctured to 2/3, ¾, 5/6 & 7/8 • Max symbol rate in Merged/Multi-Channel mode: 20MSps • Option of Enable/Disable Viterbi Decoding
Differential Decoder	• Max symbol rate – 20MSps • Option of Enable/Disable Differential Decoding • Selectable from NRZ-L, NRZ-M, NRZ-S.

OUTPUT SIGNAL AND CONNECTOR

Parameter	Specification
Supported Output Logic	• Simultaneous ECL Differential and LVDS.
ECL Differential	• Adaptive Termination • V+> -1V, V- < -1.7V
LVDS	• 100 Ohm differential Termination • Vcm = 1.2V • Amplitude – 400mVpp.
Output Connector Type	• 2 sets(channels) of 8 SMA(F) for ECL Differential • 2 RJ45 connectors for LVDS • (4 differential wire pairs are available in both case)
Input Connector Type	• 8 SMA(F) for ECL Differential

Selectable	• NRZ-L, NRZ-M, NRZ-S.
Data/Clock Signal	• BPSK (Clock and I Data – Max 20Mbps) • QSPK (Split Mode - I Data, Q Data, I Clock, Q Clock: Max 20MHz • Merged Mode – I+Q Data and Merged Clock: Max 20MSps)
Output Clock Duty Cycle	• 50% nominal and <20ppm • Holdover mode, when received signal is not present or not locked
INBUILT MODULATOR	
Parameter	**Specification**
Symbol rate	• 20KSps-20MSps
Channel	• BPSK –Single Channel, QPSK- Two Channel
Note – Detailed Modulator specification is given in separate table.	
BIT ERROR RATE MEASUREMENT	
Parameter	**Specification**
BER measuring Range	• 10^{-3} to 10^{-12} with PRBS sequence
PRBS Mode	• PRBS7, PRBS9, PRBS11, PRBS15, PRBS23, PRBS31
	• Automatic Delay adjustment between transmitting & received signal, Bit to Bit comparison between transmitted & received signal.
MONITORING AND CONTROL	
Parameter	**Specification**
Measurements	• Instantaneous display of RSSI, Eb/N0, SNR, Received Carrier Offset (Doppler), Bit Clock Offset, BER
Display	• Real Time Spectrum, Received Constellation
Control	• IF Input Settings, Demod- Settings, Bit-Sync Setting, Decoding Settings, Output Mode Setting Modulator Setting, etc.
Configuration	• Either locally through front panel GUI or remotely through TCP/IP based GUI.
Mission Mode	• Option to save configuration in file and directly loading it.

Configuration	• Either locally through front panel GUI or remotely through TCP/IP based Web GUI. • Locally, small LCD screen/Keypad in which limited configuration and monitoring is available.
OTHER SPECIFICATIONS	
Parameter	**Specification**
Reference Clock	• Support Internal as well as External clock (10 MHz)
Ethernet Ports	• Giga-bit Port over Ethernet
Size	• 19” Rack Mountable Chassis.
Operating Temperature	• 25 ± 15° C
Storage Temperature	• -10° C to 60° C
Humidity	• 20% to 80%, non-condensing
Power Supply Voltage	• 230 V ± 10%, 50 ± 3 Hz

Built-In Modulator Specifications

IF OUTPUT	
Parameter	**Specification**
No of IF Outputs	• 1, (2 ports available optionally)
Input Frequency	• 70MHz
Max Symbol Rate	• 20MSps
IF Tunability	• ±10 KHz step size
Power	• -10dBm to -40 dBm in step of 0.1dB
VSWR	• <= 1.5
IF Connector Type	• SMA Female
In Band Spurious Level	• -40dBc
IF Connector Type	• SMA Female (50 ohm)

MODULATION	
Parameter	**Specification**
Supported schemes	• BPSK, QPSK (Selectable)
Spectral Inversion support	• I&Q Swap, , &Q, I&
Symbol Rate	• 20KSps - 20MSps
Max Bit rate	• BPSK (1 X 20) Mbps • QPSK - (2X20) Mbps
Pulse Shaping Filter	• RRC with Alpha from 0.05 to 0.35 (Programmable), Programmable FIR
Modulation Enable	• Selectable (Enable/Disable)
FEC ENCODING	
Parameter	**Specification**
Convolution Encoding	• Constraint length: 7 with polynomial $(171, 133)_8$ • Selectable rate: 1/2 punctured to 2/3, 3/4, 5/6, and 7/8 • Option to enable or disable convolution encoder
Differential Encoder	• Max symbol rate – 20MSps • Option of Enable/Disable Differential Encoding
Data Source for Encoding	• External Data Source or Internal Data Source, Each Encoding Scheme can be selectively turned on or turned off.
INPUT SIGNAL AND CONNECTOR	
Parameter	**Specification**
Supported Input Logic	• Simultaneous ECL Differential and LVDS.
ECL Differential	• Adaptive Impdance; V+> -1V, V- < -1.7V
LVDS	• 100 Ohm differential Impedance • Vcm = 1.2V • Amplitude – 400mVpp.
Input Connector Type	• 2 sets(channels) of 8 SMA(F) for ECL Differential • 2 RJ45 connectors for LVDS (4 differential wire pairs are available in both case)
External Input Data Format	• Selectable from NRZ-L, NRZ-M, NRZ-S.

External Data/Clock Signal	• Both Serial and Parallel Modes Supported • BPSK (Clock and I Data – Max 20MSPS) • QSPK (Split Mode - I Data, Q Data, I Clock, Q Clock: Max 20MSps • Merged Mode – I+Q Data and Merged Clock: Max 20MSps)
Data Source to Modulator	• Internal PRBS data source for PRBS 7, 11, 15, 23, 31 • External – Data and Clock Signals in Serial and Parallel Modes on ECL or LVDS signaling.
INBUILT CHANNEL EMULATION	
Parameter	**Specification**
Symbol rate	• 10KSps - 20 MSps
Noise Source Emulation	• AWGN Noise, accurate to 6.6σ • Noise Insertion in full Sampling bandwidth of 20MHz. • Option of On/Off
Doppler Source	• Emulates Satellite Movement • Doppler shift: Up-to 1MHz • Doppler Rate Up-to 10KHz/sec. • Option of On/Off
Path Loss Insertion	• Emulation from 0dB to 50dB.
MONITORING AND CONTROL	
Parameter	**Specification**
Control	• IF Output Settings, Modulator- Settings, FEC Encoding Settings, Input Mode Setting, Modulation Setting, Channel Emulator Setting, etc.
Configuration	• Either locally through front panel GUI or remotely through TCP/IP based GUI.
Mission Mode	• Option to save configuration in file and directly loading it.
Configuration	• Either locally through front panel GUI or remotely through TCP/IP based GUI. • Locally, small LCD screen/Keypad in which limited configuration and monitoring is available.

Get in touch with us

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